ABT RECERTIFICATION EXAMINATION #22

May 6, 2005 *PART A*

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1996 and 2001. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. If you wish to return them by courier, the physical address is 9200 Sanctuary Court, Raleigh, NC 27617. An analysis of your performance in each part of the exam will be provided to the Recertification Committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please contact the ABT Office at (919) 841-5022 or abtox@mindspring.com.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.

2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.

3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

Key for the following questions
Choose the single most appropriate answer

1. Which of the following is NOT true regarding *Amanita phalloides* mushrooms?
   A) toxic components are phalloidin and amatoxins
   B) produces liver and gastroiintestinal toxicity
   C) cardiovascular toxicity is responsible for mortality
   D) common name is "death cap"
   E) no specific antidotal treatment of poisoning is available

2. Gastrointestinal tract inflammation and a swollen, yellow mottled liver are lesions that are commonly associated with ingestion of which of the following algae species?
   A) *Gymnodinium brevis*
   B) *Macrocystis pyrifera*
   C) *Microcystis aeruginosa*
   D) *Laurenia okamurai*
   E) none of the above

3. Chloroform is NOT
   A) a central nervous system depressant
   B) hepatotoxic
   C) metabolized to phosgene
   D) a peroxisome proliferator
   E) a contaminant of chlorinated water

4. The cat is more sensitive than other mammals to acetaminophen toxicity because
   A) it has accelerated bioactivation of acetominophen
   B) it is relatively deficient in glucuronyl transferase
   C) it develops severe irreversible liver toxicity
   D) it accumulates acetominophen at high levels
   E) none of the above

5. Increasing the casein content of a partially purified diet up to a level of 36%
   A) reduces the spontaneous hepatoma incidence below background in mice
   B) increases the spontaneous hepatoma incidence above background in mice
   C) has no effect on the incidence of mouse hepatomas
   D) reduces the spontaneous incidence of kidney tumors below background in mice
   E) reduces the spontaneous incidence of lung tumors below background in mice

6. All of the following may cause metabolic acidosis EXCEPT
   A) renal failure
   B) salicylates
   C) methanol
   D) diuretics
   E) diarrhea

7. A patient is admitted to the emergency room with the following symptoms: dry mouth and skin; weak, rapid pulse (130 beats/min); elevated body temperature (103 degrees F); and mydriasis. He is excited and disoriented. In his pocket is a bottle of pills labelled: "take one as necessary for stomach pain". This patient is most likely to be suffering from an overdose of
   A) a narcotic analgesic
   B) a non-narcotic analgesic
   C) an antacid
   D) an antimuscarinic agent
   E) a benzodiazepine tranquilizer

8. Each of the following solvents is paired with a correct target organ of toxicity EXCEPT
   A) methanol:retina
   B) ethylene glycol:kidney
   C) ethylene glycol monomethyl ether:kidney
   D) dichloromethane:central nervous system
   E) carbon tetrachloride:liver

9. Which of the following is NOT true about arsine?
   A) it is a gas at room temperature
   B) it produces acute intravascular hemolysis
   C) it has a garlic-like odor
   D) acute renal failure is a common manifestation of arsine poisoning
   E) significant hepatotoxicity often occurs as part of arsine poisoning

10. No specific antidote is available for poisoning by
   A) sodium fluoroacetate
   B) warfarin
   C) chlorinated hydrocarbon insecticides
   D) rotenone
   E) cyanide

11. Which of the following is NOT commonly associated with mercury vapor poisoning?
   A) acute, corrosive bronchitis
   B) interstitial pneumonitis
   C) tremor
   D) increased excitability
   E) vomiting and bloody diarrhea

12. Compounds containing beryllium are NOT readily absorbed from the gastrointestinal tract because
- A) they form insoluble phosphate precipitates at the pH of the intestinal tract
- B) they are chelated by bile salts in the small intestine
- C) their size prevents passage through cell membranes
- D) they are converted to an oxide form
- E) exposures rarely occur to the soluble metallic form

13. Which of the following is NOT associated with spermatotoxicity in rats?
- A) ethylene glycol monomethyl ether
- B) ethylene glycol monoethyl ether
- C) ethoxy acetic acid
- D) methoxy acetic acid
- E) propylene glycol monomethyl ether

14. Which form of mercury was the predominant cause of Minamata Bay disease?
- A) metallic mercury
- B) mercuric salts
- C) mercurous salts
- D) organic mercury compounds
- E) mercury was not the causative agent

15. Haber's law states that
- A) particles having an aerodynamic diameter of 1 micrometer or less will reach the alveolar region of the lung
- B) the toxic response to an inhalation exposure is a function of the concentration of the chemical multiplied by the time of exposure
- C) pulmonary edema is followed by macrophage proliferation and pulmonary fibrosis
- D) the concentration of a chemical in the blood is determined by its diffusion coefficient and plasma solubility
- E) sensory respiratory irritants cause a reduction in pulmonary compliance

16. Synthetic pyrethroids
- A) are subgrouped into type I, type II and type III
- B) alter the inactivation of voltage-gated sodium channels
- C) block glycine receptors
- D) are detoxified and eliminated slowly in mammals
- E) are primarily detoxified through conjugation with glutathione

17. Avermectins
- A) are well absorbed through skin
- B) were originally isolated from bacteria
- C) have a high affinity for chloride channels
- D) are effective against a narrow range of ecto- and endoparasites
- E) are polypeptides

18. The most important risk factor for induction of non-melanoma skin cancer is
- A) skin type
- B) exposure to solar radiation
- C) number of atypical nevi
- D) tanning booths
- E) use of sun screens

19. What is the volume occupied by 20.0 gm of hydrogen sulfide gas (MW=34) at 20˚ C and 700 mm Hg?
- A) 5.4 liters
- B) 11.3 liters
- C) 13.0 liters
- D) 15.4 liters
- E) 54 liters

20. Methyl bromide (CH_3Br)
- A) is a liquid used primarily as a fumigant
- B) has essentially no warning properties, even at physiologically hazardous concentrations
- C) is extremely flammable
- D) is of greater concern from its oral toxicity than from its inhalation toxicity
- E) would not be expected to be readily absorbed through the lungs

21. The dominant driving force in the uptake of toxic gases from the inspired air in the respiratory tract is
- A) respiration rate
- B) residual volume
- C) water/octanol partition coefficient of the toxicant
- D) alveolar ventilation
- E) diffusion

22. Which of the following radiation doses would be expected to cause symptoms of bone marrow injury, but probably not death, in people accidentally exposed to gamma radiation?
- A) 1000 rads (10 Gy)
- B) 600 rads (6 Gy)
- C) 300 rads (3 Gy)
- D) 100 rads (1 Gy)
- E) 1 rad (0.01 Gy)

23. Color additives which may be used in foods
- A) are eligible for Generally Recognized As Safe (GRAS) status
- B) include only compounds which are natural and healthful
- C) may include natural compounds which are certified by U.S. Food and Drug Administration (FDA)
- D) are either certified or exempted from certification by the FDA
- E) may be eligible for FDA approval as a food additive

24. Naturally occurring carcinogens derived from plant sources include
A) safrole
B) tobacco constituents
C) senecio alkaloids
D) aflatoxin
E) all of the above

25. The chloronicotinyl compound imidacloprid demonstrates a high insecticidal potency and exceptionally low mammalian toxicity due to
A) its high affinity for insect nicotinic acetylcholine receptors and low affinity for mammalian nicotinic acetylcholine receptors
B) the blood-brain barrier in mammals
C) the first pass effect in the liver in mammals
D) the low pH in the stomach of monogastric mammals
E) the presence of true acetylcholinesterase in mammals

26. The most commonly used pyrethroid synergist is
A) silica
B) piperonyl butoxide
C) methyl butyl ether
D) n-octyl bicyloheptene dicarboximide
E) toluene

27. A very high percentage (approximately 90%) of which type of pesticide has been shown to be carcinogenic in animal models?
A) fungicide
B) fumigant
C) rodenticide
D) insecticide
E) herbicide

28. Paraquat and diquat differ substantially in their
A) metabolism to a free radical
B) ability to initiate lipid peroxidation *in vivo*
C) uptake by the lung
D) generation of superoxide anion *in vivo*
E) mechanism of cytotoxicity

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

29. Inhibition of cytochrome P450 enzymes decreases the toxicity of which of the following?
1) benzene
2) chloroform
3) bromobenzene
4) propylene glycol

30. Which of the following statements is/are true about the toxicity of pyrethroid pesticides?
1) the mechanism of toxicity involves inhibition of cholinesterase and neurotransmitter uptake
2) the mechanism of toxicity involves ion channels regulating the passage of ions
3) piperonyl butoxide decreases the toxicity due to inhibition of the microsomal monooxygenase system
4) species sensitivity is related to efficiency of detoxification and absorption

31. Characteristic properties of solvents include(s) which of the following?
1) easy absorption via inhalation and from the gastrointestinal tract
2) the ability to cause central nervous system depression at sufficiently high acute exposure levels
3) the ability to cause dermal and mucous membrane irritation
4) the ability to cause chronic encephalopathy following acute, high level exposure

32. Deposition of particles in the respiratory tract occurs by
1) interception
2) impaction
3) sedimentation
4) diffusion
5)

33. Which of the following statements is/are true for an aerosol with a density of 1 gram/cm^3, a mass median aerodynamic diameter (MMAD) of 2.0 micrometers, and a geometric standard deviation of 2.0?
1) half of the particles are more than 2.0 micrometers in diameter
2) half of the mass of the aerosol is associated with particles less than 2.0 micrometers in diameter
3) the particle size distribution is quite narrow from 2.0 to 4.0 micrometers
4) the aerosol is polydispersed

34. Which of the following statements about metallothionein (MT) is/are correct?Which of the following statements about metallotheonein (MTs) is/are correct?
1) MTs are inducible cytosolic polymorphic proteins which have a high affinity for cadmium
2) MTs contain numerous thiol groups which allow high-affinity binding to metals
3) MTs are a factor in homeostatic control of essential metals such as zinc and copper
4) MTs bind to arsenic, lead, chromium, iron and zinc

35. Which of the following statements is/are correct?
1) stochastic effects result from cell killing with disruption of function
2) deterministic effects are seen in rapidly dividing tissues such as the gonads and hematopoietic and gastrointestinal tissues
3) deterministic effects are linear with dose and do not show a threshold for adverse effects
4) hereditary effects are of a stochastic nature and transmitted via germ cell damage

36. Which of the following solvents require(s) metabolic activation to induce toxicity?
1) benzene
2) n-hexane
3) ethylene glycol monoethyl ether
4) carbon tetrachloride

37. Which of the following is/are correct?
1) most particles larger than 10 micrometers aerodynamic diameter are deposited in the nose or oral pharynx
2) "respirable size" is generally accepted to mean particles with an aerodynamic diameter of 10 micrometers or less
3) the alveolar region has significant deposition efficiencies for particles smaller than 5 micrometers and larger than 0.003 micrometers aerodynamic diameter
4) deposition in the lung is 100% for all inspired particles of 10 micrometers aerodynamic diameter

38. Compared to adults, children are at greater risk for lead toxicity because
1) children's hand-to-mouth activity results in greater intake of lead
2) children receive a higher internal dose of lead than adults from similar exposures, because children's intestinal absorption of lead is greater and their surface area:body mass ratio is lower
3) children's exposure may occur during important periods of development
4) children generally consume more milk than adults, and the calcium in milk displaces lead from platelets

39. Which of the following is/are correct?
1) increasing the rate of respiration will increase the probability of deposition by impaction
2) increasing the rate of respiration will increase the probability of deposition in the alveolar region
3) decreasing the rate of respiration will increase the probability of deposition by sedimentation
4) decreasing the rate of respiration will increase the probability of deposition in the central airways

40. Which of the following is/are true about arsenic?
1) the primary acute toxic effect of arsine gas is hemolysis which may secondarily cause renal failure
2) the mutagenic and carcinogenic effects of arsenic are easily reproduced in gene mutation assays and in animal models
3) BAL is a primary therapeutic agent used to treat acute dermatitis and the pulmonary symptoms of acute arsenic exposure
4) symptoms of neurotoxicity are rarely seen following exposure to arsenic

41. Which of the following is/are correct regarding amygdalin?
1) it is found in the highest amounts in seeds of the bitter almond
2) the amount present in peach kernels is enough to cause poisoning in small children if several are eaten
3) ingestion of several bitter almond seeds can result in classic cyanide poisoning with death from asphyxia
4) it occurs in significant amounts in rose hips and sweet almonds

42. Ingestion of *Veratrum californicum* by ewes in early weeks of gestation results in which of the following?
1) cyclopia
2) exencephaly
3) microphthalmia
4) limb deformities

43. Cardiac effects are primary manifestations of poisoning with
1) foxglove
2) lily-of-the-valley
3) oleander
4) yew

44. Signs of copper poisoning include
1) vomiting and diarrhea
2) hypopigmentation of hair and skin
3) increased deposition of copper in liver, brain and cornea
4) macrocytic anemia

45. Features of aspirin ingestion and/or overdose include inhibition of
1) cyclooxygenase
2) synthesis of prostacyclin
3) platelet aggregration
4) synthesis of thromboxane

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PART B

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1. Which of the following is the correct xenobiotic:toxicity pair?
   A) adriamycin : tubular proteinuria
   B) polyvinyl alcohol : disseminated intravascular coagulation
   C) hydralazine : renal papillary necrosis
   D) maleic acid : mesangial fibrosis
   E) gold : glomerulonephritis

2. The primary site of kidney damage resulting from acute exposure to inorganic mercury salts is the
   A) glomerulus
   B) proximal tubule
   C) loop of Henle
   D) renal papilla
   E) entire nephron

3. The fertility index
   A) is calculated as (number of pups alive at 4 days/number of pups born alive) x 100
   B) represents the number of live fetuses per liter x 100
   C) represents the percent of copulations that result in pregnancies
   D) is calculated as (number of fetuses at 14 days/total number of implantations) x 100
   E) can be calculated as early fetal deaths/total implantations x 100

4. Which substance has been identified as a respiratory tract carcinogen in humans?
   A) kaolin
   B) hydrogen fluoride
   C) arsenic
   D) cotton dust
   E) vanadium

5. Which of the following is true?
   A) production of coagulation Factors I, II, VII, and IX is Vitamin K-dependent
   B) coagulation Factors IX and VIII are sex-linked and their genes are found on the Y chromosome
   C) activated partial thromboplastin time is a screening test of the extrinsic and common coagulation pathways
   D) prothrombin time is used to monitor warfarin therapy
   E) coagulation Factors V and XII markedly accelerate coagulation by facilitating surface attachment and localization of coagulation factors

6. Chloracne is associated with
   A) prominent hyperkeratosis of the follicular canal
   B) production of excessive sebum
   C) exposure to halogenated aliphatic hydrocarbons
   D) exposure to chlorine gas
   E) increases in serum androgen levels

7. Which of the following statements is NOT true?
   A) arsenic, benzene and vinyl chloride are known human carcinogens
   B) many peroxisome proliferators cause hepatic tumors in rats and are promoting agents for hepatocarcinogenesis
   C) benzidine, beta-naphthylamine, and derived dyes have caused urinary bladder tumors in exposed workers
   D) short asbestos fibers (<2 um long) are thought to be predominately responsible for the induction of mesotheliomas
   E) butylated hydroxyanisole (BHA) acts as a nonmutagenic carcinogen in the forestomach of rats

8. In contrast to phototoxicity, photoallergy is
   A) a Type IV delayed hypersenitivity reaction
   B) a Type II antibody dependent cytotoxic hypersensitivity reaction
   C) associated with systemic or dermal contact with a toxicant
   D) associated with exposure to UVA radiation
   E) associated with exposure to UVB radiation

9. Each of the following correctly define the terms associated with carcinogenesis EXCEPT
   A) genotoxic carcinogens interact chemically with DNA
   B) in combination syncarcinogenesis two carcinogens given one after another are additive or synergistic in effect
   C) cocarcinogens are agents that enhance the overall carcinogenic process initiated by a genotoxic carcinogen when administered before or together with the carcinogen or at a time when carcinogenic damage to DNA is still persistent
   D) promotion is a term broadly used to describe any situation yielding an increased development of neoplasms through the action of a nongenotoxic agent
   E) epigenetic carcinogens do not interact directly with genetic material

10. Ethylene glycol renal toxicity is NOT characterized by which of the following?
   A) focal hemorrhagic necrosis in the cortex
   B) metabolic acidosis
   C) accumulation of formic acid
   D) calcium oxalate crystals in the urine
   E) mild inebriation

11. Which of the following is NOT true regarding delayed hypersensitivity reactions?
- A) involves sequential recognition of specific antigen by sensitized T-lymphocytes
- B) leads to immobilization of lymphocytes at the site of sensitization
- C) involves production and release of lymphokines, which mediate increased vasodilation, accumulation of macrophages, and ultimate destruction of antigen
- D) are interactions between T- and B-lymphocytes to clear antigens
- E) is involved in the development of chronic beryllium disease

12. Which of the following assumptions is NOT correct regarding risk assessment for male reproductive effects in the absence of mechanistic data?
- A) an agent that produces an adverse reproductive effect in experimental animals is assumed to pose a potential reproductive hazard to humans
- B) in general, a non-threshold is assumed for the dose-response curve for male reproductive toxicity
- C) effects of xenobiotics on male reproduction are assumed to be similar across species unless demonstrated otherwise
- D) the most sensitive species should be used to estimate human risk
- E) reproductive processes are similar across mammalian species

13. Acidic drugs accumulate in the fetus because of which of the following?
- A) ion trapping due to reduced intracellular pH in the fetus
- B) decreased levels of alpha 1-acid glycoproteins in the fetus
- C) placental transport systems favor accumulation in the fetus
- D) fetal hepatic uptake and biliary excretion develop early in gestation
- E) fetal albumin exhibits a higher affinity than adult albumin

14. The most serious consequence of crude oil or kerosene ingestion by cattle is
- A) liver damage
- B) kidney damage
- C) aspiration pneumonia
- D) central nervous system stimulation
- E) leukemia

15. Which of the following is LEAST likely to cause hemorrhagic necrosis of the liver?
- A) beryllium
- B) dimethylnitrosamine
- C) microcystin
- D) pyrrolizidine alkaloids
- E) arsenic

16. Which region of the liver has the greatest concentration of mixed function oxidase enzymes?
- A) midzonal or zone 1 of acinus
- B) midzonal or zone 3 of acinus
- C) centrilobular or zone 1 of acinus
- D) centrilobular or zone 3 of acinus
- E) periportal

17. Toxic injury to the cell body, axon and surrounding Schwann cells of peripheral nerves are referred to, respectively, as
- A) neuropathy, axonopathy and myelopathy
- B) neuronopathy, axonopathy and myelinopathy
- C) neuropathy, axonopathy and gliosis
- D) neuronopathy, dying-back neuropathy and myelopathy
- E) chromatolysis, axonopathy and gliosis

18. Axonal transport is a key property of neural function because
- A) the volume of the axon is small relative to the volume of the cell body
- B) neurons in the adult brain have a low protein synthesis burden
- C) the axonal cytoskeleton is synthesized within the axon
- D) the primary direction of axonal transport is retrograde
- E) axonal proteins are synthesized within the neuron cell body

19. The conceptus is considered preferentially vulnerable to chemical insult due to all of the following EXCEPT
- A) rapid rate of cell proliferation
- B) requirement for precise temporal and spatial localization of cells and cell products
- C) rapid blood flow (heart rate) and increased tissue distribution of the chemical
- D) limited drug metabolism capability
- E) immaturity of the immune system

20. Which of the following is a nongenotoxic liver carcinogen in rats?
- A) aflatoxin
- B) vinyl chloride
- C) pyrrolizidine alkaloids
- D) clofibrate
- E) tamoxifen

21. Suppression of NK (natural killer) cell activity produces adverse effects in animals because
- A) NK cells play a role in the immune surveillance of tumors and the inhibition of metastases
- B) NK cells are precursors to pulmonary macrophages, and suppression of activity results in a decreased ability to combat pulmonary infections
- C) NK cells function as a helper cell in erythropoiesis, and suppression of NK cell activity results in anemia
- D) NK cells function in the complement-fixation cascade, and reduced NK cell activity causes loss of blood clotting
- E) NK cells function directly by phagocytosis of pathogens to protect the host from pathogenic bacteria (acquired cell-mediated immunity)

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

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22. Diets rich in fruits and vegetables are associated with reduced incidence of cancer. What mechanisms(s) has/have been proposed to be responsible for this?
1) inhibition of cytochrome P450 enzymes, resulting in less activation of procarcinogens to ultimate carcinogens
2) inhibition of free radical damage
3) increased activity of phase II enzymes, detoxifying activated metabolites
4) decreased bioavailability of carcinogens

23. The toxic effects of diazinon are caused by
1) covalent modification of acetylcholinesterase
2) inactivation of acetylcholine
3) accumulation of acetylcholine
4) hydrolysis of acetylcholine

24. The diversity of antibodies is controlled by
1) a separate lymphocyte for each antigen
2) gene programming following the interaction of the B-cell with antigen
3) gene programming following the interaction of the T-cell with antigen
4) the variable region of the immunoglobulin molecule

25. Which of the following is/are proposed mechanisms of non-genotoxic carcinogens?
1) chronic stimulation of cell proliferation
2) inhibition of cell death (apoptosis) in DNA-damaged cells
3) inhibition of gap junction intercellular communication
4) persistent immunosuppression

26. Chemical X induces cleft palate in offspring when administered to pregnant mice during the embryonic period. It does not induce cleft palate if it is given before implantation (although smaller litters may result), or if given after the embryonic period. Which of the following is/are correct?
1) the palate has a window of sensitivity in its development
2) during very early development, exposure to chemical X causes prenatal lethality, but not teratogenesis
3) chemical X induces cleft palate only during organogenesis
4) individual embryos are the experimental unit for statistical analysis

27. Which of the following neoplasms in humans is/are principally associated with hereditary/genetic factors?
1) osteoplastic osteosarcoma
2) xeroderma pigmentosa
3) trichoepithelioma
4) retinoblastoma

28. Which of the following is/are correct?
1) neoplasms resulting from implanted materials have only been reported in animal models
2) implanted materials that are inert do not induce neoplasia
3) the chemical nature of the implanted material is the most critical factor in determining tumorigenesis
4) the physical properties of the implanted material play a major role in determining tumorigenesis

29. A toxicant that selectively reduces follicle stimulating hormone (FSH) would be expected to do which of the following?
1) increase the production of testosterone in the testis
2) impair spermatogenesis
3) induce superovulation and thus improve fertility
4) impair recruitment and selection of the dominant ovarian follicle

30. Which of the following produces delayed neurotoxicity characterized by marked demyelination secondary to axonal changes?
1) some phosphorothioate insecticides
2) some triarylphosphates
3) some carbamate insecticides
4) some fluorine-containing alkylphosphates

31. Which of the following substances has/have been demonstrated to mediate oxidative lung tissue damage?
1) superoxide
2) nitric oxide
3) hydroxyl radicals
4) peroxynitrate

32. Which of the following is/are true?
1) diethylstilbestrol is a synthetic non-steroidal estrogen that causes vaginal adenomas in females and genital abnormalities in males exposed *in utero*
2) in rodents, retinoic acid exposure *in utero* causes targeted malformations of the central nervous system, leaving most other organ systems unaffected
3) valproic acid is used as an antiepileptic agent in pregnant women and these exposures have been associated with neural tube abnormalities
4) Bendectin® is a widely recognized human teratogen causing limb abnormalities

33. Which of the following is/are true regarding to the murine local lymph node assay (LLNA)?
1) used to predict skin sensitization potential as an alternative to the widely used guinea pig assays
2) assesses cell proliferation in the draining lymph node
3) can give a false positive response with irritants
4) total cell proliferation can be evaluated using a radioactive marker or flow cytometry

34. Type IV environmentally provoked allergic reactions are characterized by
1) immediate hypersensitivity with bronchial asthma
2) hypersensitivity pneumonitis, rheumatoid arthritis and vasculitis
3) chemically-induced hemolytic anemia and thrombocytopenia
4) cell-mediated hypersensitivity, contact dermatitis and sarcoidosis

35. Which of the following is/are function(s) of the respiratory system?
1) exchange of oxygen and carbon dioxide at the alveolar surface
2) acid-base balance through the formation of the weak acid (carbonic acid) in the blood and respiratory fluids
3) immune cell presentation via interactions with resident macrophages
4) excretion of volatile organic compounds resulting from metabolic oxidation of fatty acids

36. Which of the following is/are true regarding the blood-testis barrier?
1) transport of non-ionic chemicals is dependent upon molecular size
2) transport of non-ionic chemicals occurs via bulk flow through water-filled pores
3) transport of ionic chemicals correlates best with their degree of lipid solubility
4) the seminiferous tubules are protected the same way oocytes are protected by the blood-ovary barrier

37. Acute nephrotoxicity induced by $HgCl_2$ is characterized by which of the following?
1) glucosuria
2) aminoaciduria
3) mitochondrial dysfunction
4) reduced glomerular filtration rate

38. Alternative *in vitro* tests for ocular toxicity include
1) neutral red uptake assay
2) chorioallantoic membrane (CAM) assay
3) Tetrahymena motility assay
4) intercellular communication assay

39. A slit-lamp evaluation can determine which of the following?
1) hyperemia of the iris
2) corneal damage
3) reflex responsiveness of pupil diameter
4) thickness of the epidermis

40. Hexachlorobutadiene is
1) hepatotoxic because of a cytochrome P450 enzyme-mediated activation to a strong electrophile
2) primarily nephrotoxic in rats and mice
3) primarily an occupational hazard in the tire industry
4) nephrotoxic through the conjugation with glutathione and degradation by beta-lyase

41. Which of the following dermal reactions is/are characterized by a measurable dose-response relationship?
1) phototoxicity
2) hypopigmentation
3) primary irritation
4) contact dermatitis

42. Which of the following pairs is/are correct?
1) *N*-nitrosomorpholine : mutagenic
2) 2-acetylaminofluorene : procarcinogen
3) clofibrate : peroxisome proliferator
4) phenobarbital : nonmutagenic

43. Bromobenzene nephrotoxicity is characterized by which of the following?
1) toxic agent is a metabolite formed in extra-renal tissue
2) N-acetylcysteine S-conjugate intermediates
3) proximal tubular cell death
4) localized vasoconstriction

44. Occupational inhalation exposure to which of the following fiber types has been shown to be strongly associated with mesotheliomas and bronchogenic carcinomas?
1) rock wool fibers
2) man-made vitreous fibers
3) glass fibers
4) asbestos fibers

45. Which of the following is/are correct regarding the relative sensitivity of the thyroid gland in rats and humans?
1) long-term perturbation of the pituitary-thyroid axis by xenobiotics is more likely to cause proliferative lesions in the thyroid gland of rats
2) the greater sensitivity of the human thyroid to derangement by chemicals is related to the shorter plasma half-life of thyroxin (T4)
3) circulating T4 is bound primarily to thyroid-binding globulin in humans, but this high affinity protein is not present in adult rodents
4) a rat without a functional thyroid requires about 10 times less T4 per unit of body weight for full substitution than an adult human

46. Positive evidence of chemical carcinogenicity can include
1) increases in the number of tumors at a specific site
2) induction of rare tumors
3) earlier induction of commonly observed tumors
4) an increase in the total number of tumors

47. Which of the following is/are characteristics of a photocarcinogenicity assay in mice?
1) the assay assesses the potential of the test article to enhance ultraviolet light-induced tumors
2) a hairless strain of mouse is employed
3) a primary endpoint is the time to onset of tumors (i.e., median latent period)
4) it measures the tumorigenic response to a combination of infrared light and a chemical

48. Which of the following statements is NOT true?
1) thallium acetate at 0.1 mg/day can induce cataracts in rats within 6 weeks
2) the mechanism of galactose and other sugar-related cataracts has been explained by excessive hydration of the lens
3) posterior subcapsular opacities may result following chronic busulfan therapy
4) the mechanism of steroid-induced cataracts is believed to involve uncoupling of oxidative phosphorylation

49. The toxicity of acetaminophen is
1) due to cytochrome P450 enzyme-dependent hepatocyte activation of this drug to a reactive electrophile
2) enhanced by glutathione depletion
3) associated with centrilobular necrosis
4) not affected by alcohol consumption

50. Drugs such as purine analogues produce megaloblastic anemia by
1) destroying bone marrow stem cells
2) decreasing the absorptive capacity of the small intestine resulting in iron deficiency
3) inhibiting the synthesis of erythroid factor by the kidney
4) disturbing the maturation of erythroid cells by inhibiting DNA synthesis

ABT RECERTIFICATION EXAMINATION #22

May 6, 2005 *PART C*

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1996 and 2001. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. If you wish to return them by courier, the physical address is 9200 Sanctuary Court, Raleigh, NC 27617. An analysis of your performance in each part of the exam will be provided to the Recertification Committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please contact the ABT Office at (919) 841-5022 or abtox@mindspring.com.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.

2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.

3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

Key for the following questions
Choose the single most appropriate answer

1. Acute antihistamine poisoning from serious overdose
   A) can be easily treated with intravenous histamine
   B) rarely occurs because so few people use antihistamines
   C) can be difficult to treat due to combined central nervous system (CNS) depression and excitation resembling effects of local anesthetic overdose
   D) is characterized by CNS depression and is easily treated with CNS stimulants
   E) is characterized by CNS stimulation and is easily treated with a long-acting barbiturate

2. *In vitro* studies are typically NOT used to
   A) determine in which tissues chemical metabolism occurs
   B) determine the steps involved in the formation of metabolites
   C) discover the components of enzyme systems
   D) identify the biochemical properties of various cellular components
   E) quantify the fraction of an administered drug that is metabolized

3. Which of the following is the most significant contributor to air pollution by mass in suburban areas?
   A) manufacturing
   B) transportation
   C) space heaters
   D) electric power generation
   E) waste disposal

4. Methylation
   A) typically increases the water solubility of xenobiotics
   B) is a major pathway of xenobiotic metabolism
   C) requires S-adenosylmethionine (SAM)
   D) requires acetyl coenzyme A
   E) requires taurine

5. The mitochondrial permeability transition does NOT
   A) affect ATP synthesis
   B) contribute to necrosis
   C) contribute to apoptosis
   D) involve a proteinaceous pore
   E) result in rapid influx of Ca^{2+} into mitochondria

6. From 1973 to 1977, the average annual incidence rate of lung cancer in Connecticut was 45.9 per 100,000. The average annual prevalence was 23.0 per 100,000. From these values, what is the average duration of survival after being diagnosed with lung cancer?
   A) 2 years
   B) 6 months
   C) 1 year
   D) 8 years
   E) 4 years

7. Of the more than 40 cytochrome P450 isozymes, which 6 account for the majority of xenobiotic metabolism in humans?
   A) CYP1E1, CYP2B1, CYP2B19, CYP2F1, CYP3A7, CYP4A6
   B) CYP1A1, CYP2A6, CYP2B6, CYP2D6, CYP2F2, CYP4B2
   C) CYP1B2, CYP2C6, CYP2F2, CYP3A2, CYP3A4, CYP4A2
   D) CYP1A2, CYP2C9, CYP2C19, CYP2D6, CYP2E1, CYP3A4
   E) CYP1A1, CYP2C8, CYP2F2, CYP3A7, CYP4A2, CYP4A6

8. Which of the following conditions represents a manifestation of moderate chronic fluoride toxicity?
   A) osteomalacia
   B) osteosclerosis
   C) osteopetrosis
   D) osteopenia
   E) osteolysis

9. Intoxication from consumption of wild cherry or apricot pits would best be treated by
   A) hyperbaric oxygen
   B) artificial respiration
   C) inhalation of amyl nitrite
   D) intravenous sodium nitrite and sodium thiosulfate
   E) oral sodium nitrate

10. Drug X is available as a 2.5% solution for intravenous administration. The desired dosage of this drug is 5 mg/kg. What volume of drug should be injected if the patient weighs 50 kg?
   A) 0.2 ml
   B) 1.0 ml
   C) 2.0 ml
   D) 10 ml
   E) 20 ml

11. At present, under the federal certification program for forensic urine drug testing, analyses are performed for only five drug classes or drugs of abuse. Which of the following drugs or drug classes is NOT analyzed?
A) marijuana
B) cocaine
C) ethanol
D) amphetamines
E) opiates

12. The phrase that best defines "toxicodynamics" is the
A) linkage between exposure and dose
B) linkage between dose and response
C) dynamic nature of toxic effects among various species
D) dose range between desired biological effects and adverse health effects
E) loss of dynamic hearing range due to a toxic exposure

13. Which of the following was banned under the Delaney clause of the Food Additive Amendment of 1958?
A) butylated hydroxytoluene
B) sulfamethazine
C) cyclamate
D) phytoestrogens
E) aflatoxin

14. Flavin monooxygenases
A) are mitochondrial enzymes
B) require NADH and oxygen
C) require gamma-aminolevulinic acid
D) require coenzyme Q
E) are microsomal enzymes

15. Which of the following is NOT a characteristic of active transport?
A) blocked by saxitoxin
B) movement against a concentration gradient
C) exhibits a transport maximum
D) energy dependent
E) selectivity

16. Which of the following is NOT an initiating event in carcinogenesis?
A) DNA adduct formation
B) DNA strand breakage
C) mutation of proto-oncogenes
D) oxidative damage of DNA
E) mitogenesis

17. Which of the following does NOT inhibit electron transport?
A) rotenone
B) succinate
C) antimycin-A
D) formate
E) azide

18. Reabsorption of toxicants does NOT occur through
A) entero-hepatic recirculation
B) glomerular filtration
C) diffusion
D) active transport
E) carriers for physiologic oxyanions

19. Which of the following does NOT uncouple oxidative phosphorylation?
A) pentachlorophenol
B) dinitrophenol
C) aconitase
D) salicylate
E) gramacidin

20. Which of the following is NOT true regarding peroxisome proliferators?
A) they are a structurally diverse group of chemicals
B) they cause marked induction of lipid metabolizing enzymes
C) they often are non-genotoxic hepatocarcinogens in rodents
D) they induce hepatic CYP1A which is indicative of peroxisome proliferation
E) they operate via the peroxisome proliferator activated receptor

21. The Food Quality Protection Act of 1996 requires the Environmental Protection Agency to implement which of the following principles in its decision-making process for pesticides in/on food?
A) use of an additional ten-fold Safety Factor to protect children
B) use of the concept of "reasonable certainty of no harm" rather than "zero tolerance" of pesticides on foods that have been shown to have carcinogenic activity in animals
C) use "aggregate" exposures (combining of various routes of exposure) for risk assessment of a given pesticide
D) use "cumulative" (combined) risk assessment for chemicals that have a common mechanism of action
E) all of the above

22. Permissible Exposure Limits are
A) promulgated by the Occupational Safety and Health Administration and have the force of law
B) identical to and interchangeable with Short Term Exposure Limits
C) authorized under the Toxic Substances Control Act
D) promulgated by the Environmental Protection Agency
E) are directly adopted from the American Conference of Governmental Industrial Hygienists Threshold Limit Value list

23. Which of the following statements is NOT true regarding competitive receptor antagonist xenobiotics?
A) cause right shift in agonist dose-response curve
B) bind at the active site of the receptor
C) Emax (maximum effect) cannot be achieved
D) prevent signal transduction
E) binding at receptor is reversible

24. Spermatids and secondary spermatocytes rely heavily on which of the following metabolic pathways?
A) anaerobic metabolism
B) proteolysis
C) glycolysis
D) tricarboxylic acid (Krebs) cycle
E) lipolysis

25. Which of the following has NOT been linked to human lung cancer?
A) arsenic
B) nickel
C) silica
D) chromium
E) antimony

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

26. Which of the following is/are NOT considered a component of the standard risk assessment process?
1) hazard identification
2) dose-response assessment
3) exposure assessment
4) risk management

27. Extrapolation of developmental toxicity observed in animals to humans includes which of the following?
1) predictivity for human developmental toxicity is rabbit>monkey>rat>mouse
2) maternal toxicity that is concomitant with developmental toxicity complicates the interpretation of the studies for risk assessment
3) death, structural abnomalities, and growth alterations, but not functional deficits, are manifestations of developmental toxicity
4) in general, thresholds are assumed to exist for agents that produce developmental malformations

28. The current U.S. Environmental Protection Agency cancer risk assessment guidance
1) does not support the use of the mode of action data for assessing human relevance of tumors observed in animal studies
2) uses critical data analyses for determining options to address uncertainty
3) retains the alphanumeric scheme for characterizing the human cancer potential of agents
4) supports the derivation of reference doses (RfD) or concentrations (RfC) for chemicals producing tumors through a threshold mode of action

29. The dose response curve for an essential nutrient or vitamin
1) is U-shaped
2) is bell-shaped
3) shows adverse effects at both very low and very high doses
4) is sigmoidal

30. Factor(s) that can affect intestinal absorption and rate of absorption include
1) intestinal motility
2) drug therapy
3) gastric emptying rate
4) enterohepatic circulation

31. Chromatography techniques rely on the principle(s) of
1) disintegration of molecules into charged particles which can be separated in a magnetic field
2) atomization of the sample to the atomic state which can affect the transmittance of a designated wavelength of light
3) solubility of weak acids
4) distribution of the analyzed substance between two phases

32. Which of the following statements is/are true?
1) risk assessment includes hazard identification, dose-response assessment, exposure assessment and risk characterization
2) different regulatory agencies place different emphasis on the biological considerations used in evaluating risk
3) high to low dose extrapolations and animal to human extrapolations require numerous assumptions
4) the National Toxicology Program makes regulatory decisions based on their carcinogenicity bioassay results

33. Acceptable risk is affected by
1) the familiarity of the risk
2) media attention
3) the alternatives available
4) the reversibility of the consequences

34. Which of the following has/have the capacity for biotransformation?
1) cartilage
2) kidney
3) stratum corneum
4) plasma

35. Which of the following best describes the U.S. Interagency Coordinating Committee for the Validation of Alternative Models (ICCVAM) and the European Centre for the Validation of Alternative Methods (ECVAM)?
1) ICCVAM and ECVAM are research institutes established to develop and optimize new, non-animal test methods in toxicology
2) test methods validated by ICCVAM and/or ECVAM automatically replace the Test Guidelines of OECD, EPA and FDA
3) validation of a test method by ICCVAM and/or ECVAM is sufficient to ensure data acceptance by regulatory authorities worldwide
4) ICCVAM and ECVAM coordinate the scientific validation and promote international recognition/acceptance of alternative test methods

36. Which of the following is/are correct?
1) toxicodynamics is the quantification of absorption, distribution and metabolism of a drug
2) a key component of toxicokinetics is the measurement of elimination of the toxicant from the body
3) bioavailability is mainly determined by metabolism
4) distribution of a toxicant can depend upon its water solubility, its degree of binding to plasma proteins and its affinity for binding to tissues

37. Which of the following is/are correct?
1) plasma protein binding is mostly non-specific and reversible
2) concentration of the non-bound, active form of a drug may increase when the drug is displaced by another with more affinity for the plasma protein
3) plasma protein binding usually lengthens elimination half-life
4) basic drugs bind more readily to plasma albumin than acidic drugs

38. Which of the following describe(s) molecular transformation of compounds by intestinal flora?
1) metabolic potential of the intestine is equivalent to that of the liver
2) these transformations are mainly hydrolytic or reducing in nature
3) most placental mammals have similar microflora
4) in general, intestinal metabolism is most important for polar compounds as well as for compounds excreted free or conjugated in the bile

39. Analytical epidemiology studies include
1) case-control studies
2) case reports
3) cohort studies
4) case series

40. Which of the following statements represent(s) advantages of cohort studies over case-control studies?
1) in cohort studies, the temporal sequence between exposure and disease can be more clearly established
2) cohort studies are particularly well suited for assessing the effects of rare exposures
3) cohort studies allow for the examination of multiple effects of a single exposure
4) cohort studies are generally less expensive and time consuming

41. Acetaminophen intoxication may be treated with all EXCEPT
1) gastric lavage
2) N-acetylcysteine
3) sodium sulfate
4) phenobarbital

42. Human exposure to high concentrations of mercury vapor cause(s)
1) neurotoxicity
2) hepatotoxicity
3) nephrotoxicity
4) cardiotoxicity

43. Primary National Ambient Air Quality Standards consider which of the following?
1) protection against adverse effects
2) protection with an adequate margin of safety
3) protection of sensitive population groups
4) cost of achieving the standard

44. The U.S. Food and Drug Administration is responsible for regulating which of the following?
1) medical devices
2) environmental contaminants in foods
3) cosmetics
4) pesticide residues on raw agricultural commodities

45. Which one of the following statements is/are NOT true concerning 1,4-dioxane?
1) is a solvent with a variety of industrial uses
2) the primary metabolite in rats is b-hydroxyethoxyacetic acid
3) the primary metabolite in humans is b-hydroxyethoxyacetic acid
4) its toxicity is primarily periportal hepatocellular necrosis and nasal carcinoma in rats at high doses

46. Which of the following is/are expressions of permissable worker exposures to industrial chemicals?
1) maximum allowable concentrations
2) threshold limit values
3) short-term exposure limits
4) average daily concentrations

47. Which of the following is/are true of microsomal mixed function oxidase enzymes but NOT true of microsomal dioxygenase enzymes?
1) are membrane-bound and nonphosphorylating
2) reduce molecular oxygen to water
3) can only use NADP as an electron acceptor
4) require a second substrate to donate electrons for the reduction of an oxygen atom

48. Ozone is a significant air pollutant and health hazard because
1) it is inherently bioreactive
2) it is water soluble
3) cumulative toxicity is of greater concern than acute toxicity
4) reduction in total auto emissions in the late 1990's has significantly increased air concentrations of ozone

49. Which of the following regarding multispecies field testing is/are correct?
1) allows a more realistic exposure regime than single-species laboratory testing
2) allows investigation of a broad spectrum of environmental interactions and their effects on toxicity
3) allows the study of the influence of stressors on structural and functional properties of ecosystems
4) can be used to study compensatory mechanisms and predator-prey relationships

50. Which of the following is/are possible modes of action for a chemical to elicit an effect on pubertal development in the laboratory rat?
1) binding to a hormone receptor (antagonism or agonism)
2) interfering with hormone synthesis/metabolism (e.g., aromatase inhibitor or inhibition of iodine uptake)
3) hepatotoxicity and/or decreased weight gain due to general toxicity
4) reduced renal clearance mechanisms